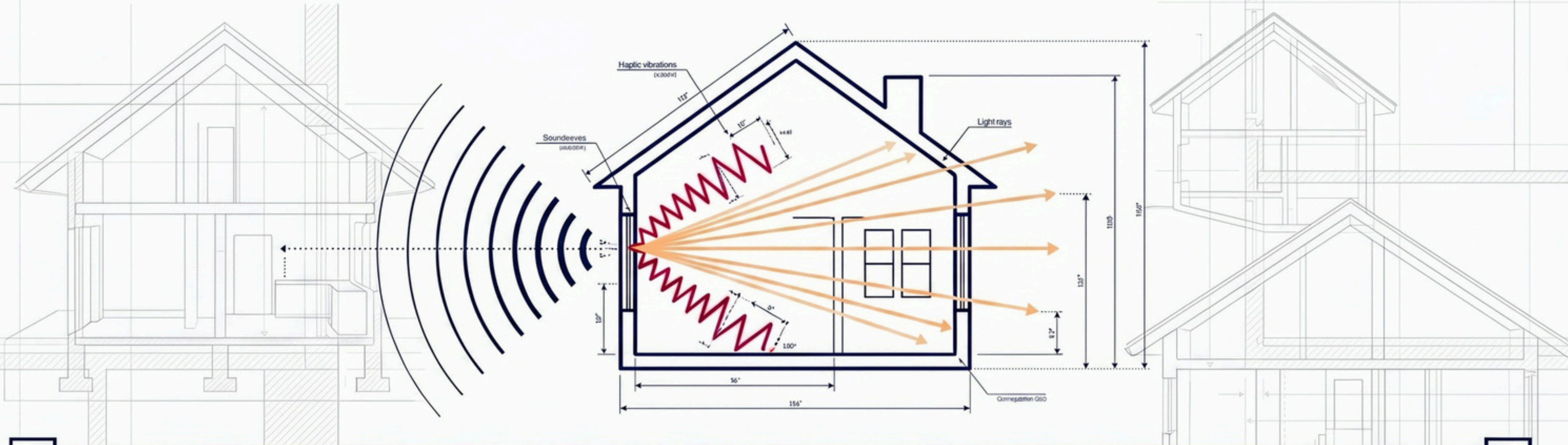


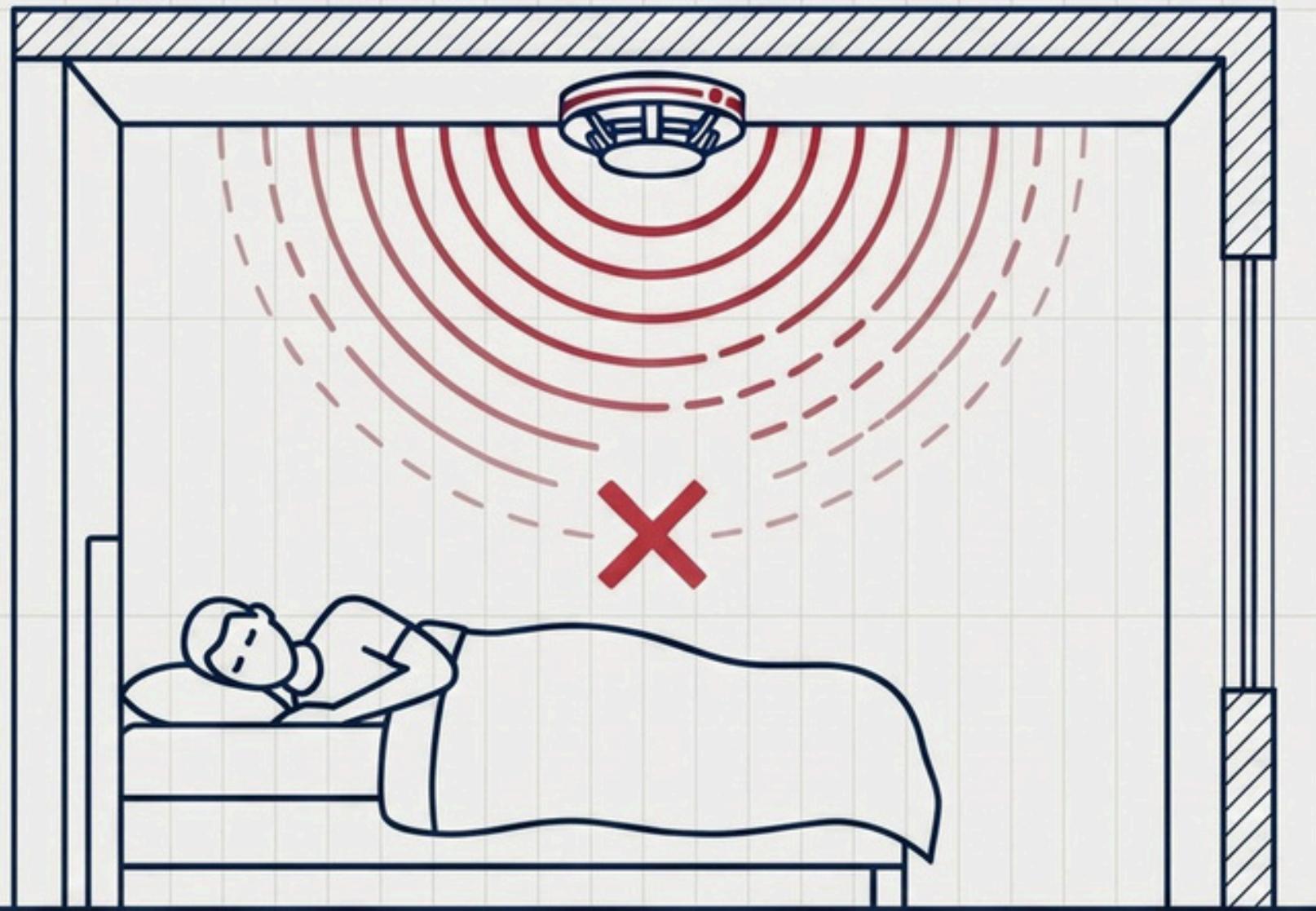
DESIGNING A FIRE ALARM SYSTEM FOR DEAF USERS



A multi-modal alert system that makes fire emergencies noticeable through vibration, visual signals, and connected home safety.

Traditional fire safety relies on sound, creating a critical vulnerability for deaf users.

The Gap



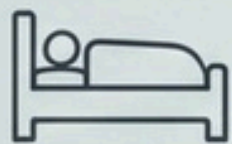
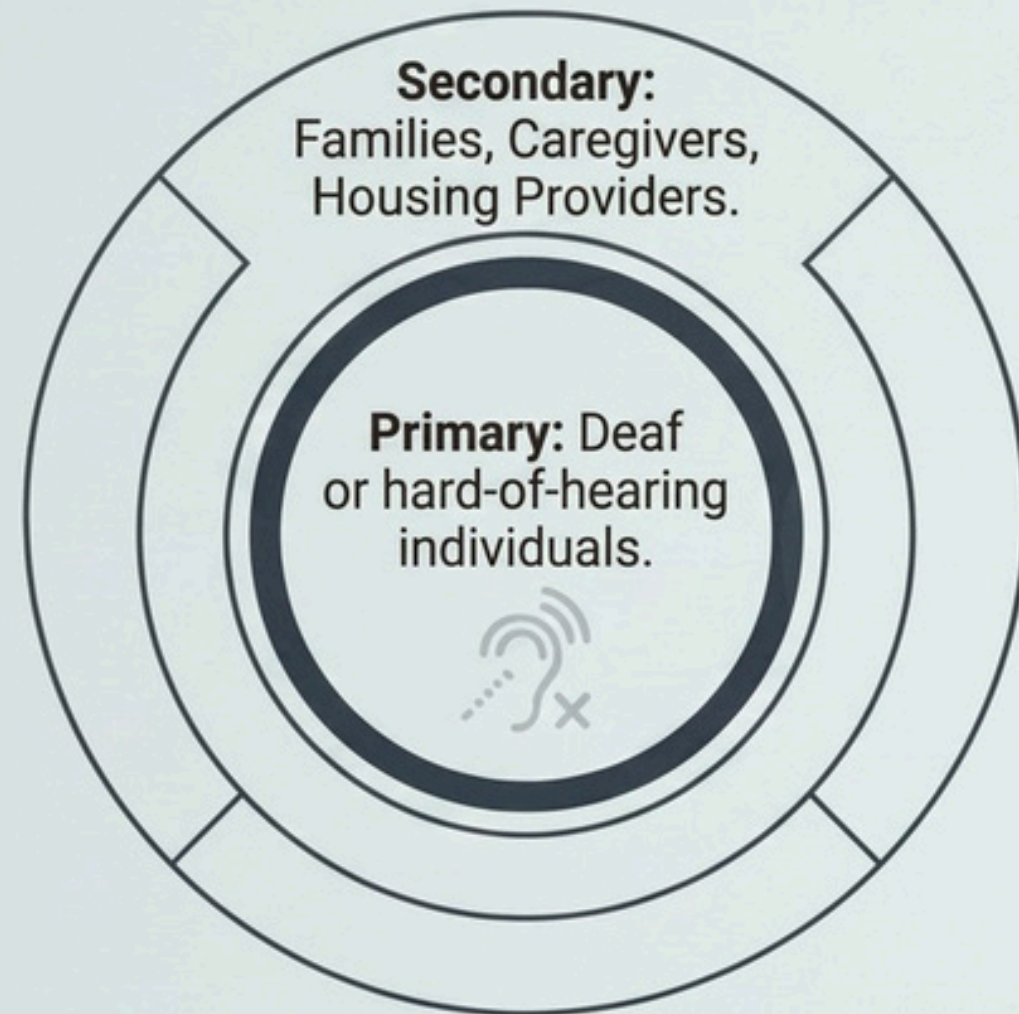
In emergencies, seconds matter. Missed alerts lead to delayed evacuation, injury, or loss of life.

The Challenge

How might we make fire alarms immediately noticeable, reliable, and easy to use for deaf users in residential environments?

Framing the scope: A physical, residential-first safety product with smart ecosystem support

Target Users & Environments



Bedrooms



Bathrooms



Kitchens



Multi-room homes

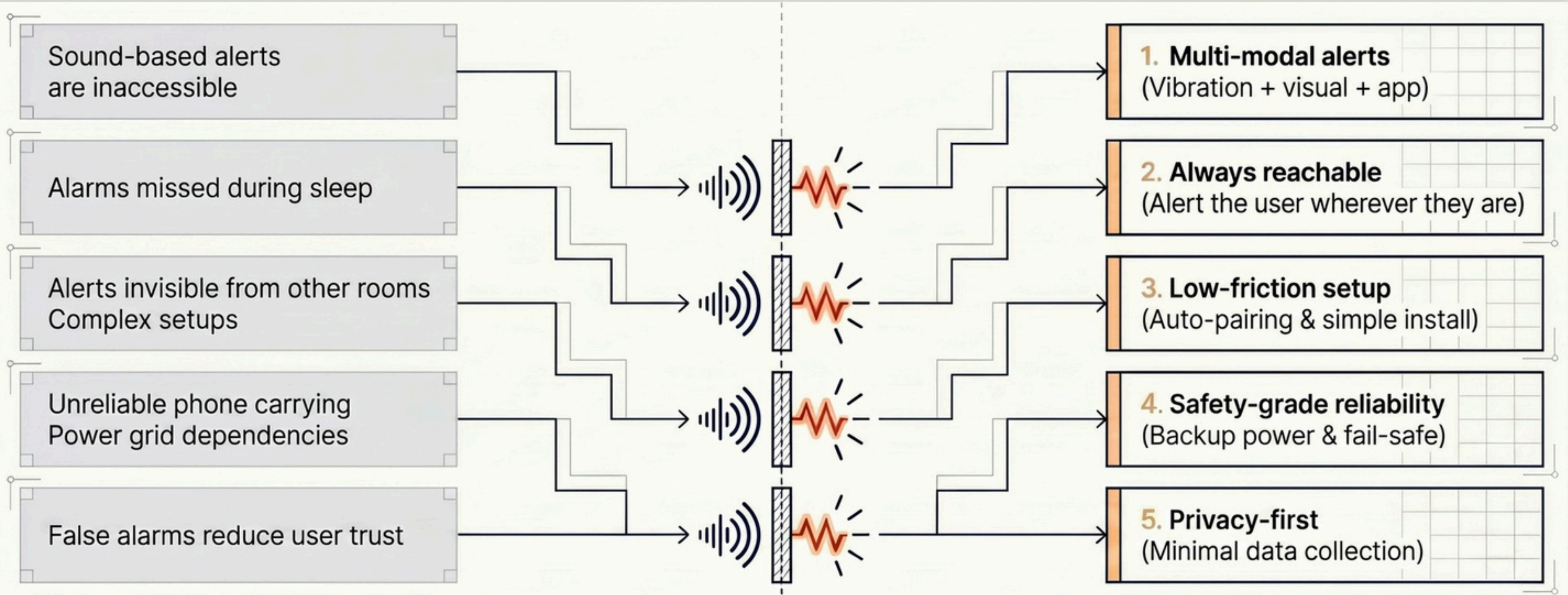


Apartments with closed doors and noisy appliances.

Scope & Constraints

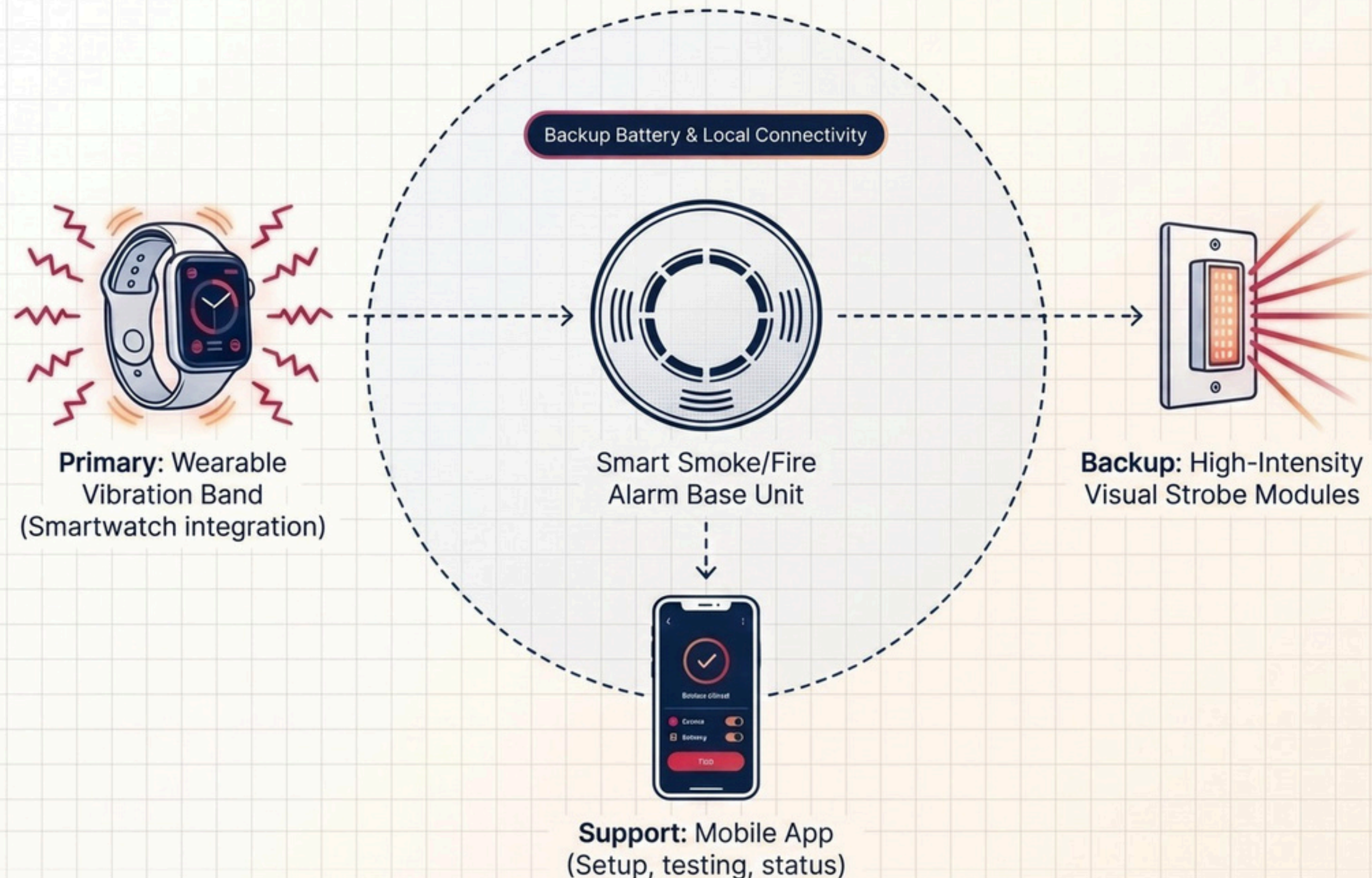
- ✓ Company manufactures home safety devices
- ✓ Focus is residential use: private homes and apartments
- ✓ Physical product is primary
- ✓ Apps and smart devices may be used as supporting technology
- ✓ Reasonable hardware budget and timeline

Every design principle must directly solve a systemic failure in the current user experience.

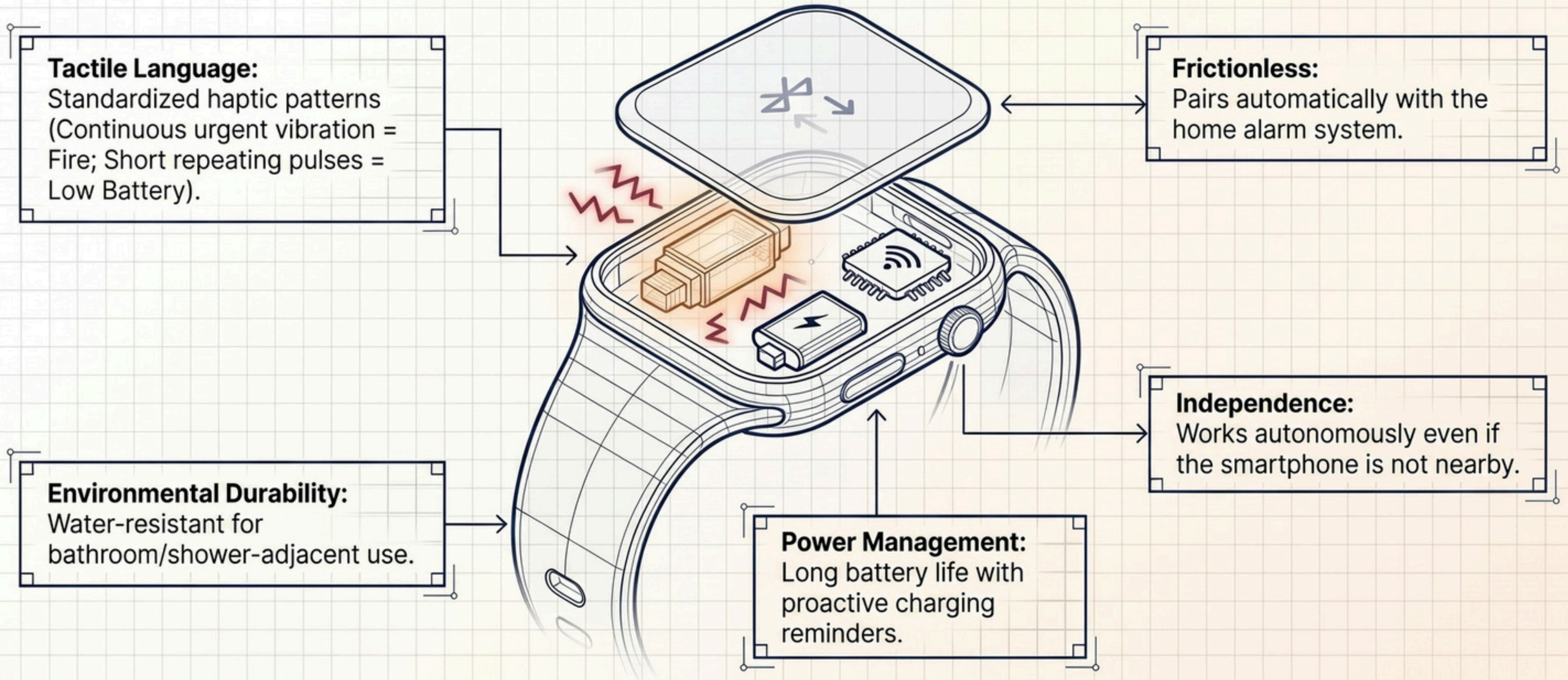


“The core need is not just detection — it is guaranteed attention.”

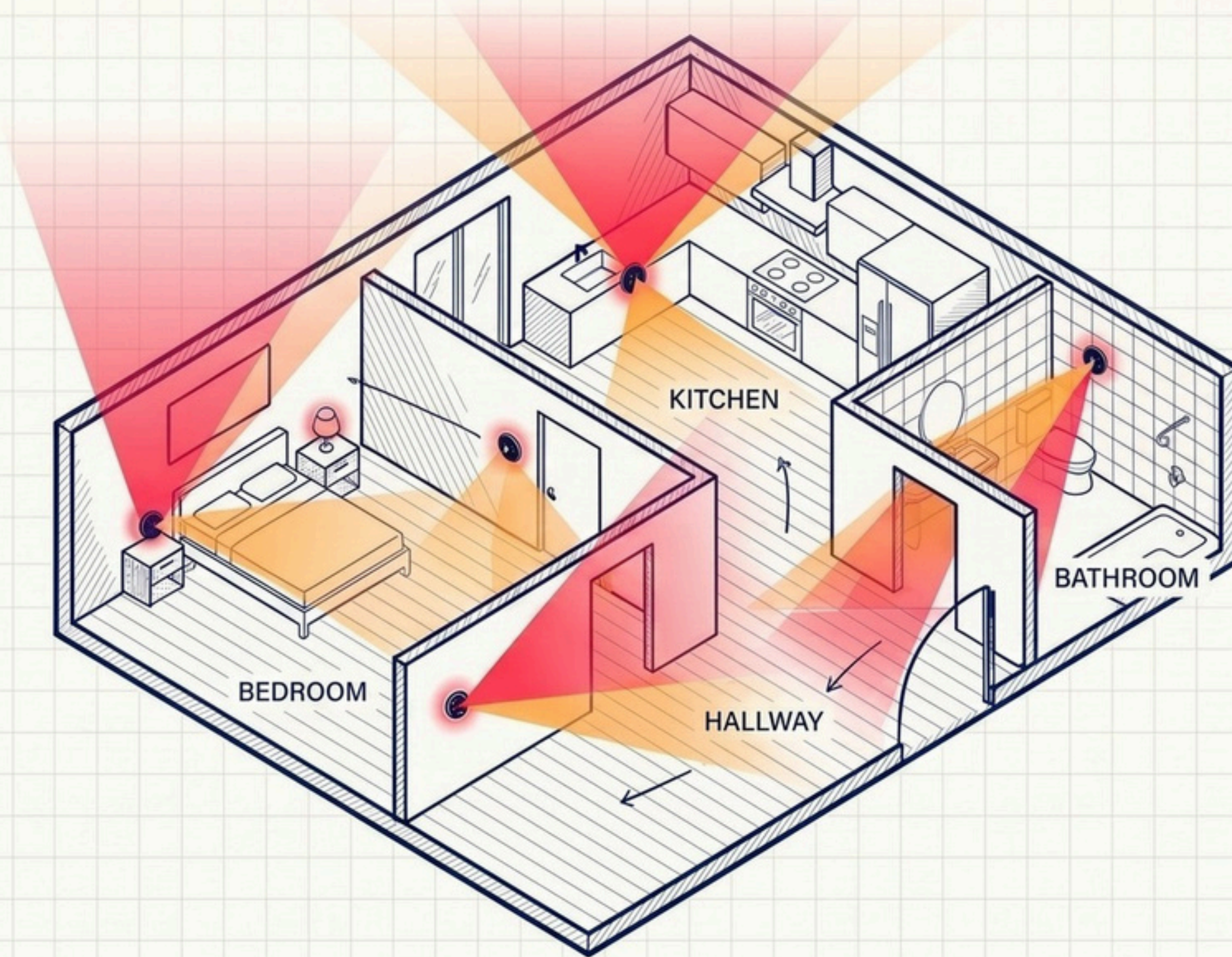
A cohesive system prioritizing tactile alerts, supported by visual redundancies and app connectivity.



The primary wearable ensures alerts are impossible to ignore, even during deep sleep or in the shower.



Room-based visual alerts provide secondary, line-of-sight redundancy across the home.

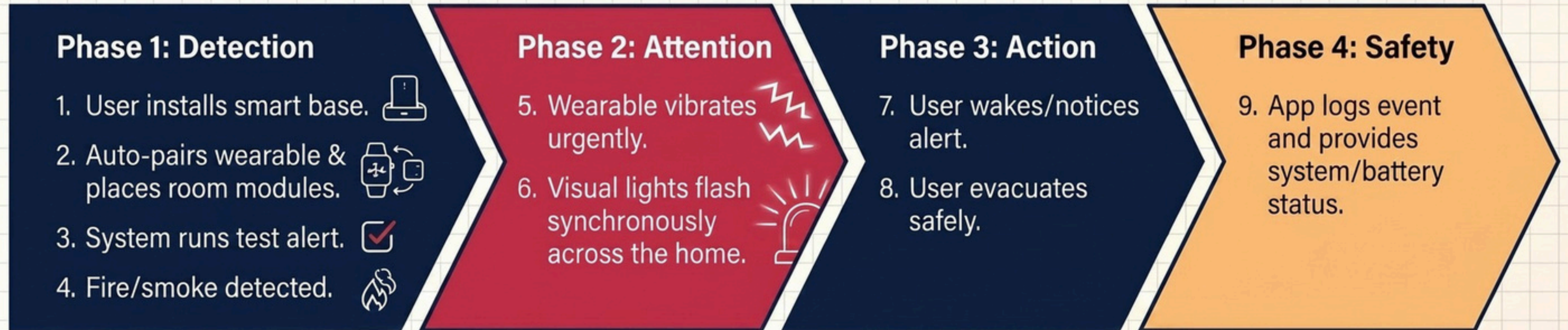


- High-intensity flashing lights positioned in key rooms.
- Distinct flash patterns distinguishing fire from device errors.
- Flexible form factors: Plug-in wall modules + optional bedside units.
- Visible through partially closed doors.

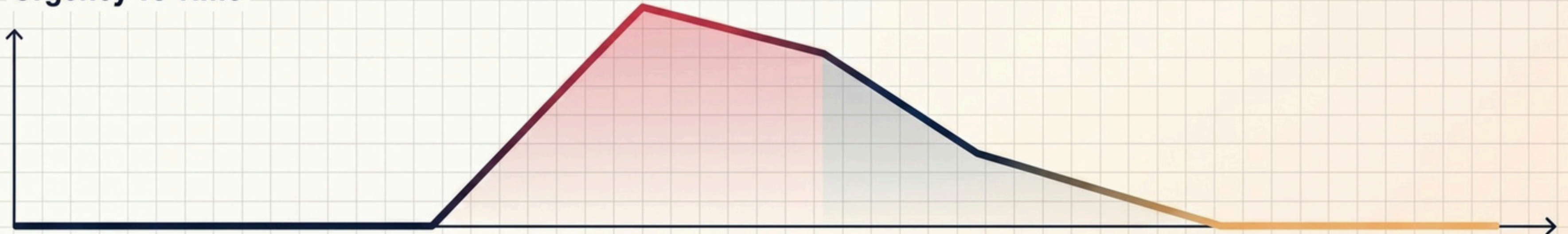
Strategic Rationale Box

Visual alerts are powerful, but users may be asleep or facing away. Therefore, they complement but do not replace tactile alerts.

From passive safety to active evacuation: mapping the critical moments of a fire event.



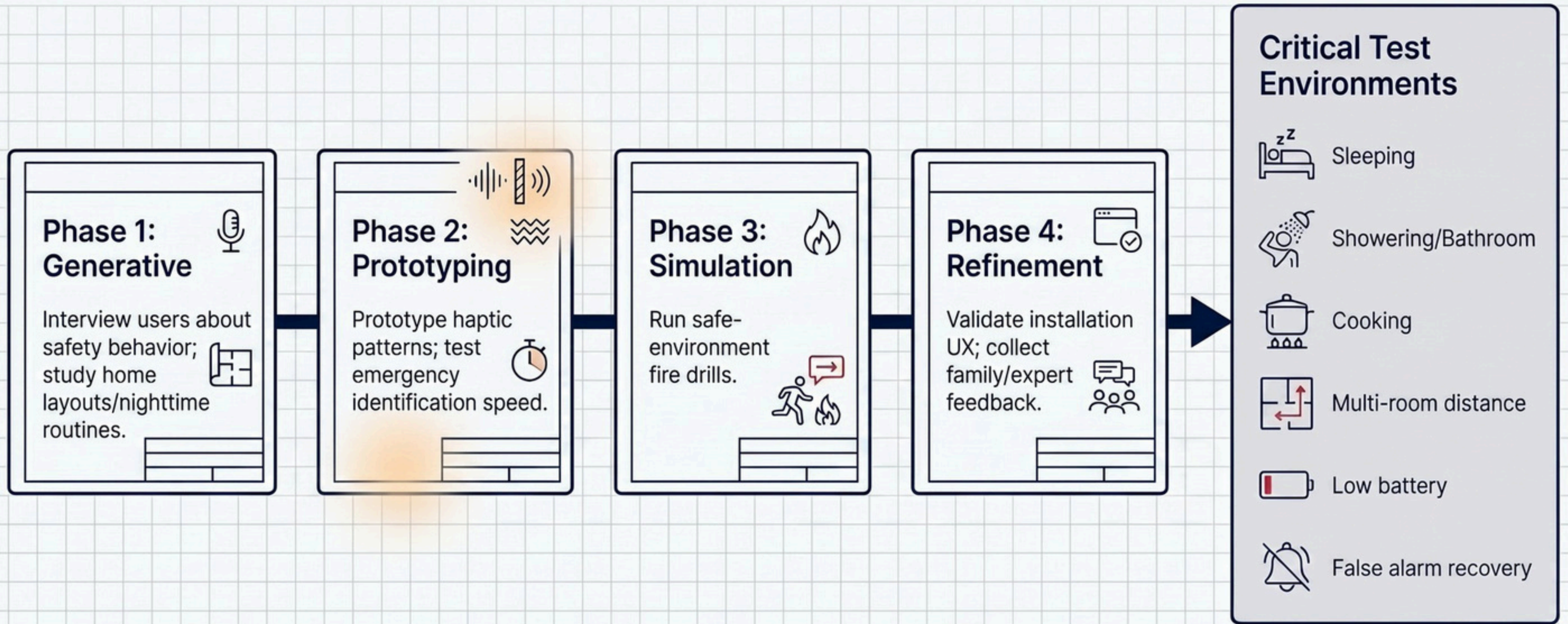
Urgency vs Time



Prioritizing tactile feedback minimizes the risk of missed alerts compared to visual or app-only methods.

Channel	Pros	Cons	Reliability During Sleep
Wearable Vibration (Primary)	Always near user, private, direct	Must be worn/charged	High 
Visual Lights (Backup)	Visible to everyone, multi-room	Missed if facing away	Medium/Low 
Mobile App / Smartwatch (Support)	Convenient, familiar	Depends on OS permissions, network, phone battery	Low 

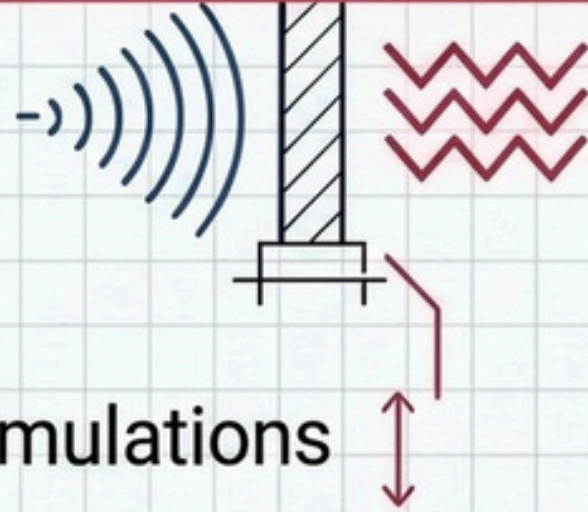
Validating life-saving technology requires rigorous, scenario-based testing with the deaf community.



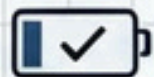
Measuring success across safety outcomes, system reliability, and business viability.

Safety Metrics

- Alert recognition rate
- Time to notice alarm 
- Evacuation action time in simulations
- Missed-alert rate 
- False alarm rate



Engagement / Reliability Metrics

- Percentage of daily wearable use 
- Battery charge compliance 
- Weekly self-test success rate
- Pairing success 
- Device uptime

Customer Metrics

- Accessibility confidence score 
- User satisfaction 
- Support tickets per install 
- Return rate
- Reviews/referrals

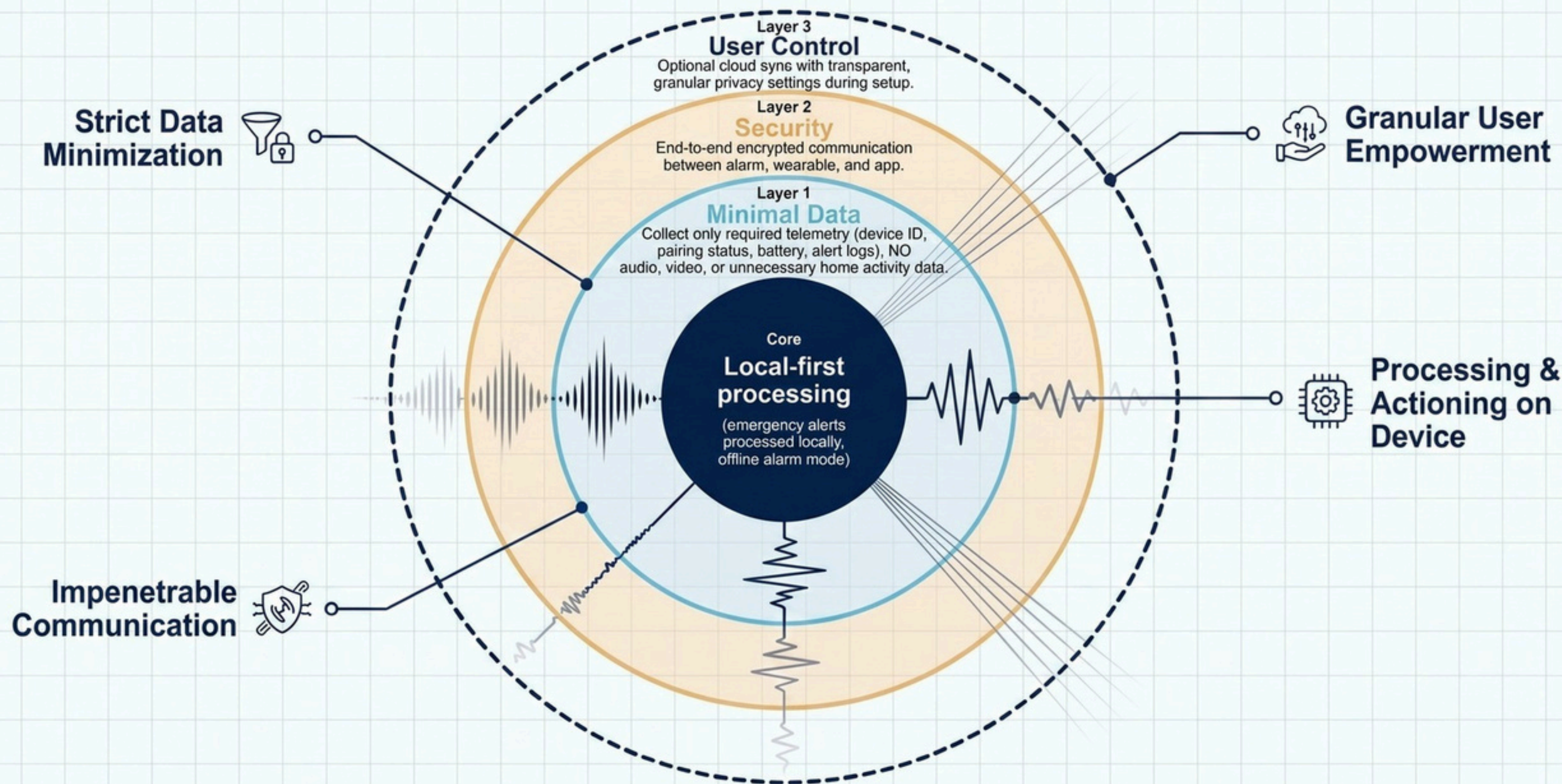


Business Metrics

- Adoption among deaf users/families 
- Partnerships with housing/accessibility orgs 
- Repeat purchases for expansion modules 



Because this is a home safety product, user trust and data privacy are as critical as functionality.



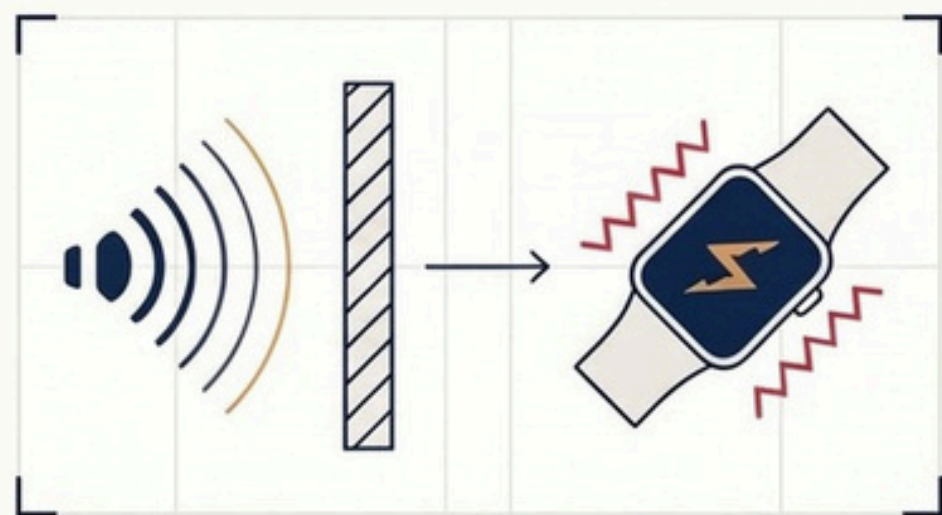
Anticipating and solving for hardware failure, human error, and edge cases.



Identified Risk	Engineered Mitigation
User forgets to wear/charge band. 	Visual backup alerts, battery reminders, bedside charging dock, weekly self-test. 
False alarms reduce trust. 	Improved sensor calibration, distinct alert patterns, easy silence/reset flow. 
Connectivity failure. 	Local radio/Bluetooth mesh network, backup battery, offline alarm mode. 
Installation friction. 	Guided app setup, QR pairing, professional installation option. 
Regulatory compliance. 	Follow fire safety and accessibility standards from early design stage. 

We don't ask deaf users to adapt to traditional alarms; we adapt the alarm to their lives.

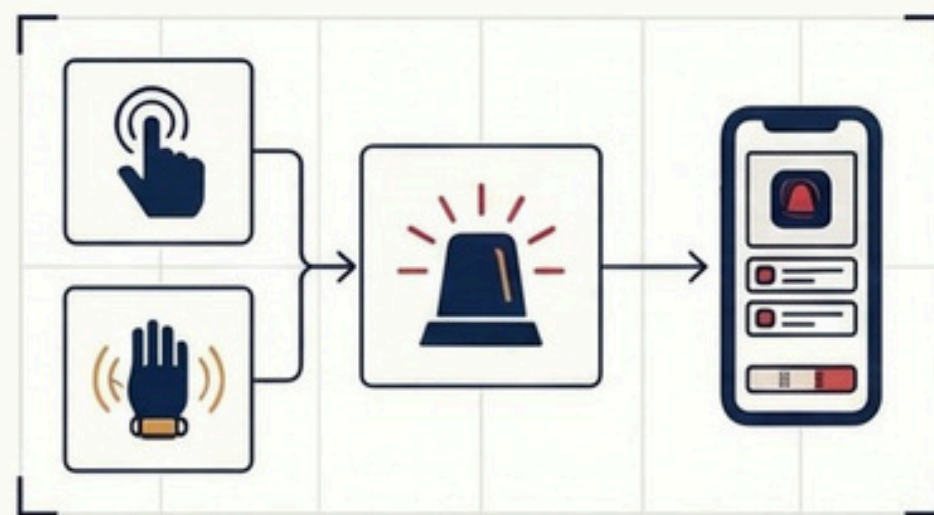
Always-Noticeable Alerts



Always-Noticeable Alerts

Wearable haptics defeat the sleep/distance barrier, ensuring detection while sleeping or showering.

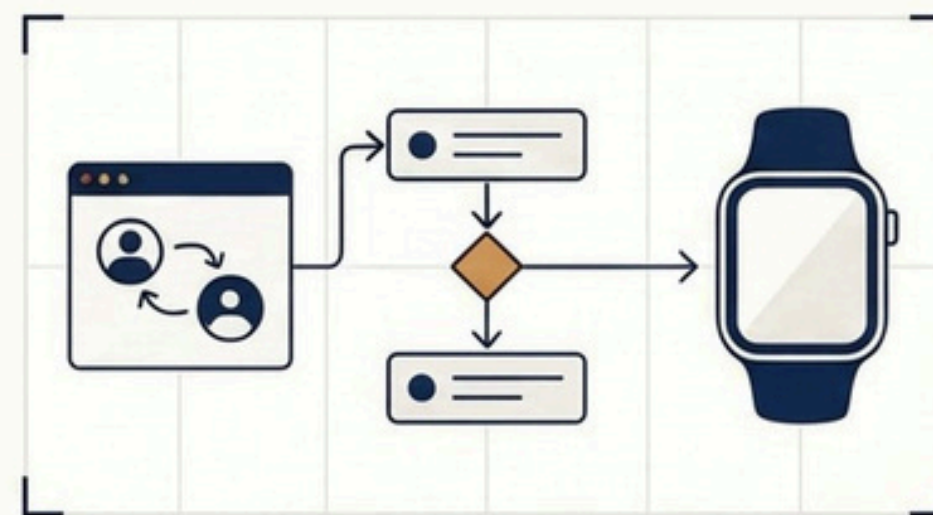
Multi-Modal Real-Home Safety



Multi-Modal Real-Home Safety

Combining tactile alerts, visual lights, and app integration eliminates single points of failure.

Low-Friction Adoption



Low-Friction Adoption

Auto-pairing, simple setup, and stylish daily-wear form factors drive consistent use.

Traditional alarms detect danger. This system ensures users actually receive the warning in time.

System Summary



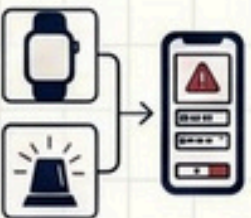
Primary user: Deaf individuals in residential homes.



Primary solution: Wearable vibration alert.



Backup solution: Room-based visual lights.



Key differentiator: Reliable, accessible, multi-modal attention.

Product design is not only about adding features — it is about identifying who is excluded by the current experience and redesigning the system around their needs... Accessibility should not be an add-on. In safety-critical products, it is the core experience.